

Prompt Basics

Zero-Shot vs Few-Shot Showdown

A comparative study of prompting techniques for customer support message classification across ChatGPT, Claude, and Gemini.

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Task: Zero-Shot vs Few-Shot Prompt Comparison

Tools tested: ChatGPT · Claude · Gemini

Dataset: 10 customer support messages

01 The Test Dataset

Ten realistic customer support messages were curated to represent three categories — Complaint, Question, and Praise — with two deliberately mixed-tone messages (#7 and #10) included to stress-test each model's judgment.

#	Message	Intended Label
1	"My order arrived 3 days late and the box was crushed."	Complaint
2	"How do I reset my password?"	Question
3	"Absolutely love this product, best purchase this year!"	Praise
4	"This is the second time my payment failed, very frustrating."	Complaint
5	"Do you ship to Pakistan?"	Question
6	"Your support team fixed my issue in 5 minutes, amazing!"	Praise
7	"I was charged twice for the same order — can someone explain why?"	Complaint
8	"What's your return policy?"	Question
9	"The app keeps crashing every time I open it."	Complaint
10	"Just wanted to say your customer service is top notch, though I wish response times were a bit faster."	Praise

Messages #7 and #10 mix sentiment intentionally — one is a complaint phrased as a question, the other is praise with a minor gripe — to test whether each approach captures overall intent rather than surface keywords.

This dataset was run through two prompting styles — zero-shot and few-shot — on three different AI tools: ChatGPT, Claude, and Gemini. The goal was to see whether providing labeled examples up front actually changes classification accuracy, and if so, by how much and on which kinds of messages the difference shows up most.

02 The Two Prompts

Zero-Shot Prompt — instruction only, no examples given:

```
Classify each of the following customer messages as "Complaint", "Question", or "Praise".  
Just give the label for each one.  
  
1. My order arrived 3 days late and the box was crushed.  
2. How do I reset my password?  
3. Absolutely love this product, best purchase this year!  
4. This is the second time my payment failed, very frustrating.  
5. Do you ship to Pakistan?  
6. Your support team fixed my issue in 5 minutes, amazing!  
7. I was charged twice for the same order — can someone explain why?  
8. What's your return policy?  
9. The app keeps crashing every time I open it.  
10. Just wanted to say your customer service is top notch, though I wish response times were a bit faster.
```

Few-Shot Prompt — instruction plus 3 labeled examples:

```
Classify each customer message as "Complaint", "Question", or "Praise".  
Use the PRIMARY tone/intent of the message, even if it's mixed.  
  
Examples:  
"My package never arrived." → Complaint  
"What time does support open?" → Question  
"You guys are amazing, thank you!" → Praise  
"I love the product, but shipping was a bit slow." → Praise (mostly positive, minor complaint)  
"Why was I charged twice? This needs fixing." → Complaint (question is secondary to the frustration)  
  
Now classify these:  
[same 10 messages as above]
```

Notice the structural difference: the zero-shot prompt gives only the task definition, leaving the model to infer on its own how to weigh conflicting signals in a message. The few-shot prompt adds five worked examples — including two that explicitly show how to resolve mixed tone — turning an implicit judgment call into a demonstrated pattern the model can follow.

03 Results — Claude

Claude was run live for both conditions. Zero-shot stumbled on the two mixed-tone messages; few-shot corrected both.

#	Zero-Shot Output	Few-Shot Output
1	Complaint	Complaint
2	Question	Question
3	Praise	Praise
4	Complaint	Complaint
5	Question	Question
6	Praise	Praise
7	Question ✗ (expected Complaint)	Complaint ✓
8	Question	Question
9	Complaint	Complaint
10	Complaint ✗ (expected Praise)	Praise ✓

Claude score: Zero-shot 8/10 | Few-shot 10/10

Both of Claude's zero-shot mistakes happened on the deliberately mixed-tone messages — #7 ("charged twice... can someone explain why?") and #10 ("top notch... though I wish response times were faster"). Without examples, the model leaned on surface cues: a trailing question mark nudged #7 toward "Question," and the word "wish" nudged #10 toward "Complaint." Once three labeled examples were added — including one that explicitly modeled how to resolve mixed sentiment — Claude correctly re-read both messages by their dominant intent rather than their surface wording, landing on a perfect 10/10.

04 Results — ChatGPT & Gemini

Both ChatGPT and Gemini were tested live in the same manner, using the identical zero-shot and few-shot prompts. Both scored a perfect 10/10 in both conditions — correctly resolving the mixed-tone messages (#7, #10) even without any examples to guide them.

#	ChatGPT — Zero-Shot	ChatGPT — Few-Shot
1	Complaint ✓	Complaint ✓
2	Question ✓	Question ✓
3	Praise ✓	Praise ✓
4	Complaint ✓	Complaint ✓
5	Question ✓	Question ✓
6	Praise ✓	Praise ✓
7	Complaint ✓	Complaint ✓
8	Question ✓	Question ✓
9	Complaint ✓	Complaint ✓
10	Praise ✓	Praise ✓

ChatGPT score: Zero-shot 10/10 | Few-shot 10/10 | **Gemini score:** Zero-shot 10/10 | Few-shot 10/10

Gemini's outputs matched ChatGPT's exactly across all 20 classifications (10 zero-shot + 10 few-shot), making it the second model to achieve a flawless run without needing examples. This suggests ChatGPT and Gemini's base instruction-following was already strong enough to infer dominant sentiment on mixed-tone messages, while Claude needed the explicit examples to reach the same level of judgment.

05 Accuracy Comparison

Tool	Zero-Shot Accuracy	Few-Shot Accuracy
ChatGPT	10/10	10/10
Claude	8/10	10/10
Gemini	10/10	10/10

06 Takeaway

Across ChatGPT, Claude, and Gemini, few-shot prompting matched or beat zero-shot in every single case — it never made results worse. ChatGPT and Gemini were strong enough to correctly resolve the two ambiguous, mixed-tone messages (#7 and #10) even with zero guidance, scoring 10/10 in both conditions. Claude, however, misread both mixed-tone messages in the zero-shot run (8/10), defaulting to surface-level cues like a trailing question mark or the word "wish" rather than judging overall intent — but jumped to a perfect 10/10 once given three labeled examples, including one that explicitly modeled how to resolve mixed sentiment. This shows that few-shot examples act as an implicit decision rule: they don't just repeat the instruction, they teach the model how to weigh conflicting signals in a message — and this benefit shows up most clearly on the model that needed it, proving few-shot prompting is most valuable exactly where a model's zero-shot judgment is weakest.